Fuel = Ethanol (C_2H_5OH)	
Oxidizer = Air (O_2 0.21 + N_2 0.79)	
Parameter	**Baseline value to note**
Wiebe n	4
Wiebe b	7
Start of combustion (SOC) (in degrees)	-14.5
Burn duration (in degrees)	45.6
Engine speed	600 RPM
Compression ratio	12
Bore	8.5 cm
Stroke	10.82 cm
Connecting rod to Crank radius Ratio	3.3
Piston Movement Option	Single Piston
Distributed Wall Heat Loss	
Coefficient a	0.1
Coefficient b	0.8
Coefficient c	0
Wall Temperature	434.0K
Woschni Correlation of Average Gas Cylinder Gas Velocity for Individual Intervals	
Compression Interval	
Coefficient C11	2.28
Coefficient C12	0.318
Coefficient C2	0.324 cm/sec-K
Ratio of Swirl Velocity to Mean Piston Speed	0
Combustion Interval	
Coefficient C11	2.28
Coefficient C12	0.318
Coefficient C2	0.324 cm/sec-K
Ratio of Swirl Velocity to Mean Piston Speed	0
Expansion Interval	
Coefficient C11	2.28
Coefficient C12	0.318
Coefficient C2	0.324 cm/sec-K
Ratio of Swirl Velocity to Mean Piston Speed	0
Combustion-Expansion Transition Temperature	1000.0K
Combustion-Expansion Transition Heat Release	0.5 cal/sec
Using Power Law for Viscosity and Thermal Conductivity	
Prandtl Number	0.7
Schimdt Number	0.7
Minimum Zone Mass	$1x10^{-6}$g
Absolute Tolerance	$1x10^{-15}$
Relative Tolerance	$1x10^{-6}$
Sensitivity Absolute Tolerance	$1x10^{-6}$
Sensitivity Relative Tolerance	0.0001

OBSERVATION			
Torque (dyne-cm)	376712600.00	378443900.00	378297000.00
Power (erg/sec)	23669550000.00	23778330000.00	23769100000.00
Thermal Efficiency	0.19	0.27	0.26

RUN	FUEL	
	$i\text{-}C_2H_5OH$	$n\text{-}C_7H_{16}$
1	0%	0%
2	100%	0%
3	80%	20%